

NAVIGATOR 1

AUTONOMOUS ROBOTICS PLATFORM | TEAM ROBONAVIGATORS

Navigator 1 is a highly fault-tolerant, AI-assisted rover powered by a custom 884-line C++ embedded operating system. Built on the Arduino UNO R4 WiFi architecture, it completely bypasses standard off-the-shelf kits. It features a standalone asynchronous web server, an active physics-based safety engine, and six intelligent driving modes, making it a truly untethered and crash-proof machine.

CORE ARCHITECTURE

Hardware & Power Systems

- **Sensor Array:** Front and rear ultrasonic sensors, a downward IR cliff sensor, and a 180° scanning servo.
- **Custom Power Architecture:** Independent 4-bay TP4056 charging station ensures perfect cell balancing.
- **Precision Telemetry:** Employs an EMA (Exponential Moving Average) filter for multimeter-accurate voltage telemetry.

Untethered Web OS

- **Zero-Lag Interface:** Custom asynchronous HTML/JS dashboard executes commands without page reloads.
- **PC-Free Boot:** A physical AP/STA toggle switch pairs with the onboard LED Matrix to scroll the active IP address, eliminating the need for a laptop monitor.
- **Network Failsafes:** A 4-second software heartbeat instantly catches dropped phone connections. An 8-second boot timeout ensures the robot safely defaults to offline autonomy if Wi-Fi is unavailable.

PHYSICS & ACTIVE SAFETY ENGINE

- **Smart Braking:** Tracks the motor state and fires a 50ms reverse-torque pulse at max PWM to instantly lock the drivetrain and kill skidding momentum.

- **Dual-Sensor Active Shield:** A 25cm virtual bumper blocks collisions in both directions, actively overriding human input if needed.
- **Cliff Protection:** Interrupt-level downward IR sensors trigger immediate braking to prevent table drop-offs.

6 INTELLIGENT OPERATING MODES

- **Mode 0 - Smart Idle Shield:** An active guard that retreats from frontal proximity, checking rear sensors first to avoid backing into walls.
- **Mode 1 - Autopilot:** Radar-assisted 180° scanning for obstacle avoidance.
- **Mode 2 - Draw & Follow:** Translates touchscreen drawings to floor vectors, with an auto-pause safety feature that freezes execution if an obstacle blocks the path.
- **Mode 3 - Object Follow:** Dynamic spatial tracking that locks onto targets (20cm–80cm).
- **Mode 4 - Voice Control:** Live, zero-latency continuous Web Speech API integration.
- **Mode 5 - Remote Control:** Precision digital joystick navigation with background crash protection.

HARDWARE WIRING & PINOUT GUIDE

PIN	CONNECTED COMPONENT	SYSTEM FUNCTION
2	Motor Driver (IN1)	Left motor direction control
3	NeoPixel Status LED	Visual diagnostic and network heartbeat indicator
4	Hardware Toggle Switch	Switches between AP Mode (Wi-Fi) and STA Mode (Hotspot)
5	Motor Driver (ENA)	Left motor speed control (PWM)
6	Motor Driver (ENB)	Right motor speed control (PWM)
7	Front Ultrasonic (ECHO)	Reads front spatial distance
8	Front Ultrasonic (TRIG)	Triggers front spatial ping
9	Servo Motor	Sweeps 180° for Autopilot and Object Follow radar
10	Motor Driver (IN2)	Left motor direction control
11	Motor Driver (IN3)	Right motor direction control
12	IR Edge Sensor	Downward-facing cliff and drop-off detection
13	Motor Driver (IN4)	Right motor direction control
A0	Analog Voltage Sensor	Reads live raw battery voltage for UI telemetry
A1	Rear Ultrasonic (ECHO)	Reads rear spatial distance
A2	Rear Ultrasonic (TRIG)	Triggers rear spatial ping